Stack

DATE

S M T W T F S

2.

i) Convert the following into postfix and show the steps similar to the examples on the slide:

a. A*B+C*D

⇒((A*B)+(C*D))

*
(
+
*
(
(

Output: AB*CD*+

Token	Stack	Output
(	(	
(	((	
A	((	A
*	((*	A
B	((*	AB
)	(	AB*
+	(+	AB*
(	(+(	AB*
C	(+(	AB*C
*	(+(*	AB*C
D	(+(*	AB*CD
)	(+	AB*CD*
)		AB*CD*+

so AB*CD*+

DATE

S M T W T F S

b、(((A+B)*C)-((D-E)*(F+G)))

~~AB+C*DE-FG+*~~	AB+C*DE-FG+*-

Token	Stack	Output
(	(	
(	((	
(	(((	
A	(((	A
+	(((+	A
B	(((+	AB
)	((	AB+
*	((*	AB+
C	((*	AB+C
)	(	AB+C*
-	(-	AB+C*
(	(-(	AB+C*
(	(-((	AB+C*
D	(-((	AB+C*D
-	(-((-	AB+C*D
E	(-((-	AB+C*DE
)	(-(	AB+C*DE-
*	(-(*	AB+C*DE-
(	(-(*(	AB+C*DE-
F	(-(*(	AB+C*DE-F
+	(-(*(+	AB+C*DE-F
G	(-(*(+	AB+C*DE-FG
)	(-(*	AB+C*DE-FG+
)	(-	AB+C*DE-FG+*
)		AB+C*DE-FG+*-

so AB+C*DE-FG+*-

DATE

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1) What is the result after evaluating the following (show the steps similar to examples in the slide

a) 78+32+/

78+32+/

3	15/5=3
~~2~~ 5	~~3+2=5~~
~~7~~ 15	7+8=15

Token	Stack	operation
7	7	push 7
8	7,8	push 8
+	15	7+8=15
3	15,3	push 3
2	15,3,2	push 2
+	15,5	3+2=5
/	3	15/5=3

b) 4 5 6 * +

34	
~~30~~	4+30=34
~~6~~	5*6=30
~~5~~	
~~4~~	

Token	Stack	operation
4	4	push 4
5	4,5	push 5
6	4,5,6	push 6
*	4,30	5*6=30
+	34	4+30=34

DATE

S M T W T F S

C, 17 10 + 3 * 9 /

9	
~~9~~	81/9=9
~~81~~	
~~3~~	
~~27~~	27*3=81
~~10~~	
~~17~~	17+10=27

Token	Stack	operation
17	17	push 17
10	17, 10	push 10
+	27	17+10=27
3	27, 3	push 3
*	81	27*3=81
9	81, 9	push 9
/	9	81/9=9